exoChile2026: Chilean Exoplanet Community Meeting

April 27-29, 2026, UCSC Concepción, Chile

Last updated: April 20, 2026

Day 1 Morning

8:30 — **Registration and arrivals**

9:15 — **Welcome & Opening Remarks**.

Overview of objectives and format of the meeting.

9:30 — **Keynote**

Andrés Jordán *Overview of Exoplanet Research from Chile*

10:10 — **Key Community Updates**. Short updates (7 min each) from different groups

- David R. Anderson *La búsqueda de exoplanetas en tránsito liderada por la UCN*
- James Jenkins *Exoplanet research from UDP/CATA*
- Rafael Brahm *The WINE collaboration: Characterizing warm jupiters to solve the hot Jupiter puzzle*
- Jeremy Tregloan-Reed *Exoplanets in Chile International collaborations and northern state telescopes*
- Alice Zurlo *Exoplanet research from YEMS*
- *Gijs Mulders / Claudia Aguilera* *Exoplanets at PUC*
- Open time for updates

11:10 — **Coffee break**

11:40 — **"Vaciado": Community Mapping.** Each (sub-)community or groups identifies:

Current strengths — Tools, facilities, and datasets — Ongoing/planned projects — Gaps, needs, opportunities and dreams

12:45 — **Group presentations**

Day 1 Afternoon

14:30 — **Lightning Talks (3 min per talk)**

- H. Parashivamurthy TESS analysis of super-Earths around super small stars
- Ricardo Ramírez Time-domain Hα spectroscopy of wide-orbit planetary-mass companions with VLT/MUSE
- Michaela Vitkova Warm gas giants and their friends
- Scarlett E Royle The effect of stellar position-velocity clustering and galactic dynamics on exoplanet properties
- Ignacia Santiago Long-Term Dynamical Stability of Multiplanet Systems' Rocky Exoplanets in the Habitable Zone
- Andrea Bernardi Spectrophotometric analysis and orbital monitoring of directly imaged substellar companions
- Alejandra Fernández Autoencoders for Speckle Denoising in ADI Sequences
- Joanne Rojas Molina RV-Astrometry Synergy in the Detection of Warm Giants
- Ilaria Giovannini Looking for satellites with doppler measurements of directly imaged exoplanets and brown dwarfs
- Bárbara Rojas-Ayala Bayesian Model Averaging for Homogeneous Stellar and Exoplanet Characterisation from Late-K to Ultracool Dwarfs

16:00 — **Coffee break**

16:30 — **Contributed talks (10 min each)**

- Felipe Lagos-Vilches Observational Constraints on the Incidence of Substellar Companions to Young Massive White Dwarfs with VLT/ERIS
- Carolina Charalambous Lithium abundance in TOI-xxx: testing the planet-engulfment hypothesis
- Connor Cheverall Ground-based Atmospheric Characterization of Volcanic Super-Earth L98-59d at High Spectral Resolution
- Adam Rains VLT/CRIRES+ observations of warm Neptune WASP-107 b: Molecular detections and challenges in ground-based transmission spectroscopy of cooler and cloudy exoplanets
- Kevin Hoy First Evidence of Radial Velocity Signal from Satellite(s) Around a Substellar Companion
- Gijs Mulders Exoplanet Water Worlds: What are they and where can we find them?
- Claudia Aguilera-Gómez Polluted White Dwarfs as Probes of Exoplanetary Systems
- Jhon Yana Galarza What Can Alpha-Enhanced Stars Tell Us About Exoplanets?
- Marcela Best Kepler in Kepler: how an "alien" astronomer would develop Kepler's Third Law

Day 1 Evening

18:00 — **Community Icebreaker** (food/drinks)

19:00 — Public talk "Exoplanetas y vida en el espacio" by Alice Zurlo at Universidad Católica de la Santísima Concepción

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Day 2 Morning

9:30 — **Recap & Inspiration Talks.** Review of Day 1 + short "provocations" talks about:

- E-ELT opportunities and other coming facilities
- new surveys
- AI for exoplanets

10:15 — **Mixed-Community Dialogues.** Participants rearrange into **diverse mixed groups**. Each group discusses:

- Collaboration opportunities
- Chile's role in global exoplanet science
- Barriers to collaboration

11:15 — **Coffee Break**

11:50 — **Plenary Report-Back (~1 hr)**

- Groups summarize key ideas
- Cluster outputs on a "community map" (wall or screen)
- Emergent themes guide the afternoon's seed-lab topics

Day 2 Afternoon

14:00 — **Lightning talks (3 min each)**

- Fernando Castillo M. — Survival of vortices in gas-rich debris disks
- Erick Montoya — Unveiling the real geometry of the FU Orionis system
- Constanza Norambuena Herrera — Polluting white dwarfs: a dynamical study
- Alex Mauch-Soriano — Predicted incidence of Jupiter-like planets around white dwarfs
- Sidharth Aroli Veettil — The 14N/15N and 12C/13C isotopic ratios in the protoplanetary disk around V4046 Sgr
- Constanza Castillo — The formation of formaldehyde (H2CO) in protoplanetary disks
- Marie M. Rodríguez S. — Variability and Geometry of the Brγ Emission in HD 100546 from VLTI/GRAVITY
- Milenne Avila Bravo — Planetary architectures under the influence of a stellar binary
- Lucas Gonzalez-Rivas — Not Just Gas: How Solid-Driven Torques Shaped the Migration of the Galilean Moons
- Grace Batalla Falcon — Evolution of the silicate feature in protoplanetary disks
- Camilo González-Ruilova — Can the environment affect the planet formation?
- Hannah Osborne — A homogenous analysis of small-planet masses

14:40 — **Contributed talks (10 min each)**

- Nicola Astudillo-Defru — Searching for terrestrial exoplanets with optimized radial velocities
- Carlos Ferreira — Beyond BLS: A New Approach to Exoplanet Period Detection
- Claudio Caceres — A search for free-floating planets in nearby young stellar associations
- Neda Hejazi — Exploring Planetary Formation and Composition Using Detailed Chemical Abundances of Host Stars
- Philipp Weber — Planet formation in the light of stellar accretion bursts
- Matthias Schreiber — The importance of tides for the survival of planets during late stellar evolution

15:40 — **Collaboration Seed-Labs.** Self-organized breakout groups form around concrete ideas. For example:

- Joint proposals
- Shared data/AI tools or open repositories
- Training programs or student exchanges
- Outreach and public engagement initiatives

16:20 — **Coffee Break**

16:45 — **Seed-Lab Pitches (5 min/each).** Groups present concrete next steps and timelines.

17:15 — **Community Strategy Session** (Roadmap Discussion). Possibly as a small panel

- National priorities
- Future facilities
- Student training pathways

Day 2 Evening

19:00 — Public talk "Del Desierto de Atacama al cosmos" by Bárbara Rojas-Ayala at Universidad Católica de la Santísima Concepción

Day 3 Morning

9:45 — **Student-led reflection session.** A session *organized by students, moderated by postdocs.*

- Training needs
- Mentoring networks
- What barriers do students face in Chile?
- Community structure (how can the community better support early-career researchers?)
- Guided reflections on the need of national summer schools on exoplanet research, statistics and machine learning, or other topics
- Chilean exoplanet Slack workspace

10:45 — **From ideas to Working Groups**

11:00 — **Coffee Break**

11:30 — **Working Groups and Commitments.** Here's where we consolidate initiatives from seed labs and other sessions.

- Organizers compile “exoChile 2026 Community Initiatives”

12:00 — **Community Roadmap Discussion**

12:45 — **Next Steps and Closing Remarks**

- Who hosts exoChile2027
- Creation of communication channels

13:00 — **Farewell and Group Photo**

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List of Posters

ATM-01	Benjamin Gonzalez Gonzalez Vidal	Modeling of Archean Earth-like aquaplanet atmosphere: the effect of obliquity on climate and observables of exoplanets
ATM-02	Catalina Valesca Ávalos Vega	Breaking Degeneracies in Exoplanet Atmospheres: Measuring Rayleigh Slopes in exoplanets to Infer Mean Molecular Weight
DET-01	Sebastián Elias Cabrera Astudillo	BODIES: Big Or Dinky Investigation and Evaluation System
DET-02	Marcela Cristina López Ulloa	When M Dwarfs Flicker: Are Young Exoplanets Hiding There?
DET-03	Daniela Gonzalez	The Synergy of Gaia DR4 and VLTI/GRAVITY+: Low-Mass exoplanet detection in Binary Systems
DET-04	Joanne Michelle Rojas Molina	RV-Astrometry Synergy in the Detection of Warm Giants
DET-05	Florian Rodler	Mass determination of the three long-period Neptune- and sub-Neptune-sized planets transiting TOI-282
DET-06	Cathalina Andrea Zamora Fernández	Discovery and characterization of a long period system
DET-07	Franshezka Dominik Paz Tapia Candia	The discovery and characterization of a long period system
DET-08	José Luis Ricra Mayorca	Orbital period refinement and analysis of central transit time variations of WASP-50 b using ground-based and space observations
DET-09	Jose Vines	TIC 275647095 b, an eccentric Warm Jupiter Orbiting a Solar-analog Star
DET-10	Maximiliano Moyano	Discovery and characterization of a long period system
DISK-01	Hernán René Salinas Castro	Searching for Complex Organic Molecules with JWST–MIRI in Planetesimal Formation Regions
DISK-02	Abigali Rodríguez Jiménez	Rotation curves of prototype protoplanetary discs
EDU-01	Sol Francisca Troncoso Barrías; Lindsay Rincón Torres; Thiare Antonella Molina Tapia	Introduction to Data Science through the Exploration of Exoplanets as an Educational Strategy
INST-01	Oscar Castillo Vega	An optimized implementation of the PACO algorithm.
INST-02	Ian Rickmers Blamey	Deepermoments: Deep Learning for moment maps
OUT-01	Martina Pascall Molina Riffo; Bárbara Scarlett Quilodrán Hijerra; Javiera Cajas Reyes	Astronomical Collage as a Science Communication Tool with a Gender Perspective in Exoplanet Science Outreach
PLAS-01	Misael Sánchez Alexander Rebolledo	Modeling collective plasma dynamics in space using Hamiltonian mean-field approaches
SYS-01	Felipe Ignacio Rojas Henríquez	Giant companions in Super-Earth systems from TESS
SYS-02	Álvaro Ignacio Llancaqueo Albornoz	Unveiling the nature of TESS giant planets in the first PLATO Field
SYS-03	Bhavya Ambalapatta kannancheri	Is the Hot Neptune Desert Really Empty? A Comparison of TESS and Kepler.

Poster Code Legend (Numbers indicate individual posters within each thematic category.)

DET — Detection & Characterization of Exoplanets
ATM — Exoplanet Atmospheres & Climate
SYS — Planetary Systems & Demographics
DISK — Protoplanetary Disks & Planet Formation
INST — Instrumentation, Methods & Algorithms
DS — Data Science, Machine Learning & Education
OUT — Outreach, Communication & Society
PLASMA — Plasma Physics & Fundamental Processes